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Control apparatus including driving unit, lens apparatus, image pickup apparatus, control method, and storage medium

Abstract

A control apparatus for a camera system that includes an image pickup apparatus and a lens apparatus including a driven unit, an output unit that is supplied with power from the image pickup apparatus, an actuator configured to move the driven unit includes a driving unit configured to drive the actuator by a PWM output method or a linear output method using the power from the output unit, and a control unit configured to control an output of the power from the output unit to the driven unit. The control unit causes the output unit to output a first power in a case where the driving unit drives the actuator by the PWM output method, and in a case where the driving unit drives the actuator by the linear output method, the control unit causes the output unit to output a second power different from the first power.

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Background/Summary

BACKGROUND
Technical Field
(1) One of the aspects of the embodiments relates to a control apparatus, a lens apparatus, an image pickup apparatus, a control method, and a storage medium.
Description of Related Art
(2) The number of pixels and the sensitivity of an image sensor have recently been increased in an image pickup apparatus, and the influence of noise generated by a pulse width modulation (PWM) control for controlling an actuator has become remarkable.
(3) Japanese Patent Laid-Open No. 2003-149525 discloses a configuration for controlling an actuator by an amplifier output method that does not generate noise.
(4) However, the power efficiency of the amplifier output method is lower than that of PWM control, and thus the number of images is reduced in the configuration of Japanese Patent Laid-Open No. 2003-149525.
SUMMARY

(5) A control apparatus according to one aspect of the embodiment is used for a camera system that includes an image pickup apparatus and a lens apparatus including a driven unit, an output unit that is supplied with power from the image pickup apparatus and outputs power that is used to move the driven unit, and an actuator configured to move the driven unit. The control apparatus includes a driving unit configured to drive the actuator by a PWM output method or a linear output method using the power from the output unit, and a control unit configured to control an output of the power from the output unit to the driven unit. The control unit causes the output unit to output a first power in a case where the driving unit drives the actuator by the PWM output method, and in a case where the driving unit drives the actuator by the linear output method, the control unit causes the output unit to output a second power different from the first power. A lens apparatus and an image pickup apparatus each having the above control apparatus, a control method corresponding to the above control apparatus, a storage medium storing a program that causes a computer to execute the above control method also constitute another aspect of the disclosure.
(6) Further features of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of a camera system according to one embodiment of the disclosure.
(2) FIG. **2** is a detailed diagram of a control apparatus.
(3) FIG. **3** is a flowchart illustrating an example of power setting processing.
(4) FIG. **4** is a flowchart illustrating another example of power setting processing.
(5) FIG. **5** is a flowchart illustrating another example of power setting processing.
(6) FIG. **6** is a flowchart illustrating another example of power setting processing.
(7) FIG. **7** is a flowchart illustrating another example of power setting processing.
DESCRIPTION OF THE EMBODIMENTS
(8) In the following, the term “unit” may refer to a software context, a hardware context, or a combination of software and hardware contexts. In the software context, the term “unit” refers to a functionality, an application, a software module, a function, a routine, a set of instructions, or a program that can be executed by a programmable processor such as a microprocessor, a central processing unit (CPU), or a specially designed programmable device or controller. A memory contains instructions or programs that, when executed by the CPU, cause the CPU to perform operations corresponding to units or functions. In the hardware context, the term “unit” refers to a hardware element, a circuit, an assembly, a physical structure, a system, a module, or a subsystem. Depending on the specific embodiment, the term “unit” may include mechanical, optical, or electrical components, or any combination of them. The term “unit” may include active (e.g., transistors) or passive (e.g., capacitor) components. The term “unit” may include semiconductor devices having a substrate and other layers of materials having various concentrations of conductivity. It may include a CPU or a programmable processor that can execute a program stored in a memory to perform specified functions. The term “unit” may include logic elements (e.g., AND, OR) implemented by transistor circuits or any other switching circuits. In the combination of software and hardware contexts, the term “unit” or “circuit” refers to any combination of the software and hardware contexts as described above. In addition, the term “element,” “assembly,” “component,” or “device” may also refer to “circuit” with or without integration with packaging materials.
(9) Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure. Corresponding elements in respective figures will be designated by the same reference numerals, and a duplicate description thereof will be omitted.

(10) FIG. **1** is a block diagram of a lens interchangeable type single-lens digital camera system, which is an example of a camera system (optical apparatus) **100** according to one embodiment of the present disclosure. The camera system **100** includes a camera body (image pickup apparatus) **150** and an interchangeable lens (lens apparatus) **101** attachable to and detachable from the camera body **150**. However, the present disclosure is not limited to this example, and can also be applied to a camera system **100** in which an image pickup apparatus and a lens apparatus are integrated with each other. In the camera system **100** in which the image pickup apparatus and the lens apparatus are integrated, the camera CPU has a function of the lens CPU.
(11) In the camera system **100**, the camera body **150** and the interchangeable lens **101** are connected via a communication terminal (communication contact portion) **153** in a camera-side contact portion **152** and a communication terminal (communication contact portion) **104** in a lens-side contact portion **103** so that they can communicate information with each other. (Electric) power is supplied from a power circuit unit **155** in the camera body **150** to a power circuit unit (output unit) **106** of the interchangeable lens **101** via a power contact portion **154** in the camera-side contact portion **152** and a power contact portion **105** in the lens-side contact portion **103**. The power circuit unit **155** generates various power sources used in the camera body **150** or power sources to be supplied to the interchangeable lens **101** from a battery **156** attached to the camera body **150** using a Low Dropout (LDO) (linear regulator) or a DC-DC circuit.
(12) A lens CPU **102** as a lens controller provided in the interchangeable lens **101** stores characteristic information and optical information unique to the interchangeable lens **101** in an internal memory **140**. The lens CPU **102** stores in the internal memory **140** a set value to be output to a driver IC provided in each of a focus lens driving circuit **112**, a zoom lens driving circuit **126**, an aperture driving circuit **115**, and an image stabilization (IS) driving circuit **116**. However, the present disclosure is not limited to this example, and the set values may be stored in a memory provided separately from the lens CPU **102**. The lens CPU **102** transmits the characteristic information and the optical information to a camera CPU **151** as a camera controller provided inside the camera body **150** via the communication terminals **104** and **153**.
(13) The characteristic information includes a name of the interchangeable lens **101** (ID information for identifying the model), a maximum communication speed, maximum F-number, whether it is a zoom lens or not, a compatible autofocus (AF) system, and an image height capable of AF. The characteristic information further includes information about table data indicating a relationship between an F-number and a T-number. The optical information includes sensitivity information about the focus lens **107** obtained by a matrix of a position of a focus lens **107**, a position of a zoom lens (magnification varying lens) **108**, and a state of an aperture stop **109**, a focus correction amount (designed value), a focus correction manufacturing error value.
(14) A permission signal or the like for permitting movement of the focus lens **107** by operating a manual focus (MF) driving user interface (UI) **122** provided in the interchangeable lens **101** is also transmitted from the camera CPU **151** to the lens CPU **102**. The interchangeable lens **101** and the camera body **150** exchange information such as other operation states, setting states, various information request commands (transmission requests), and driving commands via the communication terminals **104** and **153**.
(15) The interchangeable lens **101** includes a lens UI **120** such as a switch for selecting AF or MF in the focusing operation. The state of the lens UI **120** is also exchanged via the communication terminals **104** and **153**.
(16) The interchangeable lens **101** includes an optical system (imaging optical system) including the focus lens **107**, the zoom lens **108**, the aperture stop **109**, and an image stabilizing lens **110**. The zoom lens **108** and focus lens **107** are each movable in a direction along an optical axis OA (optical axis direction). The image stabilizing lens **110** is movable in a direction intersecting the optical axis OA. A light beam from an object formed via the imaging optical system is guided to an image sensor **162** provided within the camera body **150**. As a method for detecting the focus state, the

image sensor **162** can output a phase difference signal and an image signal at the same time if including a structure in which one pixel has a plurality of photoelectric conversion units.
(17) In a case where the camera CPU **151** confirms that AF is selected by the AF/MF selection switch included in the lens UI **120**, it starts AF operation. The camera CPU **151** processes the output from the image sensor **162**, detect the focus state of the imaging optical system, combines it with the above optical information of the interchangeable lens **101**, and determines a moving amount of the focus lens **107** to obtain an in-focus state for the object.
(18) The camera CPU **151** transmits the calculated focus lens moving amount to the lens CPU **102** via the communication terminals **104** and **153**. The lens CPU **102** controls the focus lens driving circuit **112** in accordance with focus lens position information from a position sensor **111** that detects the position of the focus lens **107** and the received moving amount of the focus lens **107**. Thereby, the focus lens **107** can be moved to the in-focus position.
(19) The user can operate the MF driving UI **122** including a cylindrical manual focus ring that rotates around the optical axis OA and is mounted on the exterior of the interchangeable lens **101**. A detection sensor **123** detects an operation amount of the MF driving UI **122** by the user. The lens CPU **102** controls the focus lens driving circuit **112** according to the operation amount of the MF driving UI **122** to move the focus lens **107** to a predetermined position. In a case where the camera CPU **151** confirms that MF is selected by the AF/MF selection switch, the camera CPU **151** performs focusing by moving the focus lens **107** to a predetermined position according to the user operation of the MF driving UI **122** without performing the AF operation.
(20) A user can operate a zoom driving UI **124**. The zoom driving UI is, for example, a cylindrical manual zoom ring mounted on the exterior of the interchangeable lens **101** rotatably around the optical axis OA. A detection sensor **125** detects an operation amount of the zoom driving UI **124**. The lens CPU **102** controls the zoom lens driving circuit **126** based on the operation amount of the zoom driving UI **124** and the detection signal of the position detection sensor **127** that detects the position of the zoom lens **108**, and moves the zoom lens **108** to a predetermined position. The zoom driving UI may be provided on the camera body **150**.
(21) The camera CPU **151** determines the result of photometry by a photometry sensor (not illustrated) based on a half-press operation of a release switch included in a camera UI (camera user interface) **161** provided on the camera body **150**. The camera CPU **151** determines an F-number (aperture value) set by operating an operation unit included in the camera UI **161**.
(22) The camera CPU **151** transmits an F-number set for the aperture stop **109** to the lens CPU **102** via the communication terminals **104** and **153**. The lens CPU **102** moves the aperture stop **109** (controls the aperture of the aperture stop **109**) by controlling the aperture driving circuit **115** according to the received F-number and aperture stop position information based on a position sensor **114** that detects the position of the aperture stop **109**.
(23) The camera CPU **151** transmits an image stabilization start command to the lens CPU **102** via the communication terminals **104** and **153** in response to the half-pressing operation of the release switch. Upon receiving the image stabilization start command, the lens CPU **102** first controls the IS driving circuit **116** to hold the image stabilizing lens **110** at a control center position. Next, the lens CPU **102** controls a lock driving circuit **117** to drive a mechanical lock **118** to release a locked state. Thereafter, the lens CPU **102** controls the IS driving circuit **116** according to the detection result of the camera shake detecting circuit **119** to move the image stabilizing lens **110** for image stabilization.
(24) The camera CPU **151** moves a main mirror (not illustrated) and the shutter **163** installed in front of the image sensor **162** according to the full-pressing operation of the release switch, guides a light beam from the imaging optical system to the image sensor **162**, and captures an image. The image sensor **162** includes a photoelectric conversion element such as a CCD sensor or a CMOS sensor. The camera CPU **151** generates image data based on an output from the image sensor **162** and records it on a recording medium. In a case where a still image capturing mode is selected by a

setting of a camera GUI unit including the display unit **164** and a camera UI **165**, an image to be captured is a still image. In a case where a moving image capturing mode is selected by the setting of the camera GUI unit, an image to be captured is a moving image. Alternatively, a moving image recording start button may be separately provided, and a moving image recording may be started in a case where the recording start button is pressed. A user can check a captured image through an electronic viewfinder **166**.
(25) FIG. **2** is a detailed diagram of the control apparatus **200**. The control apparatus **200** includes the lens CPU **102** and the IS driving circuit **116**. The IS driving circuit **116** includes a driver IC (driving control unit) **128** and an actuator **129**. In this embodiment, the image stabilizing lens **110** is moved as a driven unit, but this embodiment is not limited to this example. The driven unit in another embodiment may be, for example, the aperture stop **109** or the focus lens **107**.
(26) The actuator **129** includes moving coils **131** and **132** attached to the image stabilizing lens **110** and magnets (permanent magnets) **133** and **134** disposed to face the moving coils **131** and **132**, respectively. The image stabilizing lens **110** is fixed together with the moving coils **131** and **132** to an unillustrated fixed frame. By applying a current to the moving coils **131** and **132**, a driving force can be generated between the moving coils **131** and **132** and the magnets **133** and **134**, and the image stabilizing lens **110** can be moved.
(27) The driver IC **128** includes a PWM output circuit and a linear output circuit, and can switch the output method (PWM output method and linear output method) to the moving coils **131** and **132** by communication from the lens CPU **102**. The output method may be switched by changing the setting of the terminal of the driver IC **128** from the lens CPU **102** instead of the communication from the lens CPU **102**.
(28) The power circuit unit **106** supplies power to the driver IC **128**. The power circuit unit **106** outputs power (second power) in an output method in which a step-down DC-DC circuit that receives the power supplied from the power circuit unit **155** via the power contact portions **154** and **105** outputs the power. The power circuit unit **106** can output power (first power) in an output method in which the power supplied from the power circuit unit **155** via the power contact portions **154** and **105** is output as it is. An analog-to-digital (A/D) converter (detector) **141** of the lens CPU **102** detects a voltage level supplied from the power circuit unit **155** via voltage dividing resistors. The step-down DC-DC circuit can turn on and off the output and set the output voltage level by communication from the lens CPU **102** or terminal setting. The control unit **142** controls the output method of the power circuit unit **106**. A step-up DCDC circuit or a step-up/step-down DCDC circuit may be used instead of the step-down DC-DC circuit.
(29) The operation of the IS driving circuit **116** will be described below. FIG. **3** is a flowchart illustrating an example of power setting processing.
(30) In step S**101**, the lens CPU **102** determines whether the output method of the driver IC **128** is set to the PWM output method. In a case where the lens CPU **102** determines that the PWM output method is set, the flow proceeds to step S**102**; otherwise (in a case where the linear output method is set), the flow proceeds to step S**103**.
(31) In step S**102**, the lens CPU **102** causes the power circuit unit **106** to supply the power supplied from the power circuit unit **155** to the driver IC **128** as it is.
(32) In step S**103**, the lens CPU **102** enables the DC-DC output of the power circuit unit **106** and causes the power circuit unit **106** to supply power to the driver IC **128** through the DC-DC output.
(33) In a case where the linear output (output by amplifier) format is set, the flow in FIG. **3** can efficiently reduce the voltage using the DC-DC output to the maximum power required for the output. Since there is a power loss even if the voltage is stepped down by the DC-DC output, in a case where the PWM output method is set, directly supplying power to the driver IC **128** enables the PWM control to efficiently use power. Therefore, efficient power supply is available for each output method of the driver IC **128**.
(34) FIG. **4** is a flowchart illustrating another example of power setting processing.

(35) In step S**201**, the lens CPU **102** determines whether the voltage level supplied from the power circuit unit **155** detected by the AD converter **141** is higher than a predetermined value. In a case where the lens CPU **102** determines that the voltage level is higher than the predetermined value, the flow proceeds to step S**203**; otherwise (in a case where the voltage level is less than the predetermined value), the flow proceeds to step S**202**. In a case where the voltage level is equal to the predetermined value, the flow may proceed to any one of them.
(36) In step S**202**, the lens CPU **102** determines whether the current output of the power circuit unit **106** is set to the DC-DC output. In a case where the output of the power circuit unit **106** is set to the DC-DC output, the flow proceeds to step S**204**; otherwise, the flow proceeds to step S**203**.
(37) In step S**203**, the lens CPU **102** maintains (does not change) the output method of the power circuit unit **106**.
(38) In step S**204**, the lens CPU **102** changes the output of the power circuit unit **106** from the DC-DC output to an output method in which the power supplied from the power circuit unit **155** is supplied to the driver IC **128** as it is.
(39) As illustrated in FIG. **5**, the order of processing steps S**201** and S**202** in FIG. **4** may be reversed. In FIG. **5**, in step S**301**, the lens CPU **102** determines whether the current output of the power circuit unit **106** is set to the DC-DC output. In a case where the output of the power circuit unit **106** is set to the DC-DC output, the flow proceeds to step S**302**; otherwise, the flow proceeds to step S**304**. In step S**302**, the lens CPU **102** determines whether the voltage level supplied from the power circuit unit **155** detected by the AD converter **141** is lower than a predetermined value. In a case where the lens CPU **102** determines that the voltage level is lower than the predetermined value, the flow proceeds to step S**303**; otherwise (in a case where the voltage level is higher than the predetermined value), the flow proceeds to step S**304**. In a case where the voltage level is equal to the predetermined value, the flow may proceed to any one of them. In step S**303**, the lens CPU **102** changes the output method of the power circuit unit **106** from the DC-DC output to the output method in which the power supplied from the power circuit unit **155** is supplied to the driver IC **128** as it is. In step S**304**, the lens CPU **102** maintains (does not change) the output method of the power circuit unit **106**.
(40) By executing the flow of FIG. **4** or **5**, in a case where the voltage level supplied from the power circuit unit **155** is lower than the predetermined value and the power supply output method is the DC-DC output, a difference between input and output potentials reduces, the efficiency decreases, and power loss can be avoided. In addition, the reduced difference between input and output potentials can suppress noises caused by the DC-DC control such as a change in the DC-DC switching period and switching skips.
(41) FIG. **6** is a flowchart illustrating another example of power supply setting.
(42) In step S**401**, the lens CPU **102** determines whether the interchangeable lens **101** is in the power saving operation state. The power saving operation state is set, for example, in a case where each actuator of the interchangeable lens **101** is not operated for a certain period of time due to long-exposure imaging, or in a case where the power consumption of the camera system **100** as a whole is large due to attachment of an accessory, or the like. The power saving operation state is also set in a case where the image stabilizing operation is turned off and the image stabilizing lens **110** is held at the center of the optical axis, or in a case where the aperture stop **109** is held at a predetermined aperture. The power saving operation state is also set in a case where the power saving of the focus lens **107** is maintained. In a case where the lens CPU **102** determines that the actuator **129** is in the power saving operation state, the flow proceeds to step S**402**; otherwise, the flow proceeds to step S**404**.
(43) In step S**402**, the lens CPU **102** determines whether the current output of the power circuit unit **106** is set to the DC-DC output. In a case where the output of the power circuit unit **106** is set to the DC-DC output, the flow proceeds to step S**403**; otherwise, the flow proceeds to step S**404**.
(44) In step S**403**, the lens CPU **102** lowers the set value of the DC-DC output. That is, the lens

CPU **102** causes the power circuit unit **106** to output a third power having a voltage different from (lower than in this embodiment) the second power.
(45) In step S**404**, the lens CPU **102** maintains (does not change) the output method of the power circuit unit **106**.
(46) As illustrated in FIG. **7**, the order of processing steps S**401** and S**402** in FIG. **6** may be reversed. In FIG. **7**, in step S**501**, the lens CPU **102** determines whether the current output of the power circuit unit **106** is set to the DC-DC output. In a case where the output of the power circuit unit **106** is set to the DC-DC output, the flow proceeds to step S**502**; otherwise, the flow proceeds to step S**504**. In step S**502**, the lens CPU **102** determines whether the actuator **129** is in the power saving operation state. In a case where the lens CPU **102** determines that the actuator **129** is in the power saving operation state, the flow proceeds to step S**503**; otherwise, the flow proceeds to step S**504**. In step S**503**, the lens CPU **102** decreases the DC-DC output setting value. In step S**504**, the lens CPU **102** maintains (does not change) the output method of the power circuit unit **106**.
(47) By executing the flow of FIG. **6** or **7**, in a case where the voltage level required for actuator operation changes, the power can be efficiently reduced to the required level by the DC-DC output, and the power efficiency can be improved.
Other Embodiments
(48) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disc (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.
(49) This embodiment can provide a control apparatus, a lens apparatus, an image pickup apparatus, a control method, and a storage medium, each of which can suppress the influence of noise without reducing the power efficiency.
(50) While the disclosure has been described with reference to embodiments, it is to be understood that the disclosure is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(51) This application claims the benefit of Japanese Patent Application No. 2022-129454, filed on Aug. 16, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A control apparatus for a camera system that includes an image pickup apparatus and a lens apparatus including a driven unit, an output unit that is supplied with power from the image pickup apparatus and outputs power that is used to move the driven unit, and an actuator configured to

move the driven unit, the control apparatus comprising: a driving unit configured to drive the actuator by a PWM output method or a linear output method using the power from the output unit; and a control unit configured to control an output of the power from the output unit to the driven unit, wherein the control unit causes the output unit to output a first power in a case where the driving unit drives the actuator by the PWM output method, and in a case where the driving unit drives the actuator by the linear output method, the control unit causes the output unit to output a second power different from the first power, and wherein in a case where the output unit outputs the second power and the lens apparatus is in a specific state, the control unit causes the output unit to output a third power having a voltage different from a voltage of the second power to the output unit.

2. The control apparatus according to claim 1, wherein the first power is the power supplied from the image pickup apparatus, and wherein the second power is power with a voltage different from the voltage of the power supplied from the image pickup apparatus.

3. The control apparatus according to claim 2, wherein the output unit includes a DC-DC circuit, and wherein the second power is power output from the DC-DC circuit to which the power supplied from the image pickup apparatus is input.

4. The control apparatus according to claim 1, further comprising a detector configured to detect a voltage level supplied from the image pickup apparatus to the output unit, and wherein the control unit causes the output unit to output the first power in a case where the output unit outputs the second power and the voltage level is lower than a predetermined value.

5. A lens apparatus comprising: the control apparatus according to claim 1; a driven unit; an output unit supplied with power from the image pickup apparatus and configured to output power that is used to move the driven unit; and an actuator configured to move the driven unit.

6. An image pickup apparatus comprising: the control apparatus according to claim 1; and an image sensor.

7. A control method for a camera system that includes an image pickup apparatus and a lens apparatus including a driven unit, an output unit that is supplied with power from the image pickup apparatus and outputs power that is used to move the driven unit, and a driving unit for driving an actuator, the actuator being configured to move the driven unit, the control method comprising the steps of: driving, by the driving unit, the actuator by a PWM output method or a linear output method using the power from the output unit; and causing the output unit to output a first power in a case where the driving unit drives the actuator by the PWM output method, and in a case where the driving unit drives the actuator by the linear output method, causing the output unit to output a second power different from the first power, wherein in a case where the output unit outputs the second power and the lens apparatus is in a specific state, the output unit is caused to output a third power having a voltage different from a voltage of the second power to the output unit.

8. A non-transitory computer-readable medium storing a program that causes a computer to execute the control method according to claim 7.